Name: Aarav Sharma | USN / Roll No: 1PH23CS001

Subject: Engineering Physics - IA 1 | Section: A

Ans 1)

Newton's first law: A body continues in its state of rest or of uniform motion in a straight line unless an external force acts on it. This is also called the law of inertia.

Second law: Rate of change of momentum is directly proportional to the applied force and is in the direction of the force. F = ma.

Third law: For every action there is an equal and opposite reaction. Eg: when a gun is fired the gun recoils backwards.

Ans 2)

Given u = 0, v = 20 m/s, t = 5 s

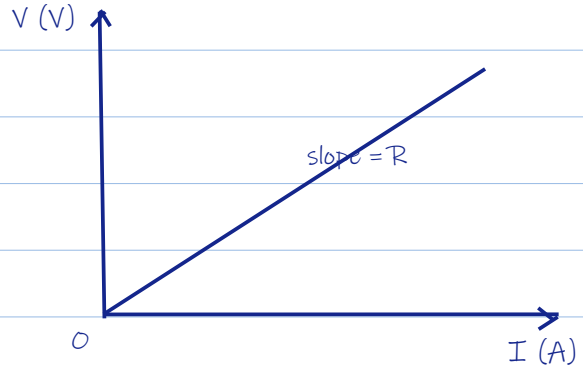
(a) a = (v - u)/t = (20 - 0)/5 = 4 m/s^2

(b) s = ut + 1/2 at^2 = 0 + 1/2 x 4 x 5^2 = 50 m

Distance = 50 m

Ans 3)

Ohm's law: At constant temperature, the current flowing through a conductor is directly proportional to the potential difference across its ends. $V = IR$ where R is the resistance.



The graph is a straight line passing through the origin. Resistance $R = V/I$ = slope of the V-I graph, which is constant for an ohmic conductor.

Ans 4)

KE is the energy due to motion, $KE = 1/2\ mv^2$. PE is the energy due to position (height), $PE = mgh$.

By conservation of energy, $mgh = 1/2\ mv^2$

$v = \sqrt{2gh} = \sqrt{2 \times 10 \times 20} = \sqrt{400} = 20$ m/s

Name: Diya Patel | USN / Roll No: 1PH23CS014

Subject: Engineering Physics - IA 1 | Section: A

1.

First law - an object at rest stays at rest and an object in motion keeps moving with the same velocity unless a force acts on it (inertia).

Second law - force is equal to mass times acceleration, F = ma.

Third law - every action has an equal and opposite reaction. Example: a rocket pushes gases down and the gases push the rocket up.

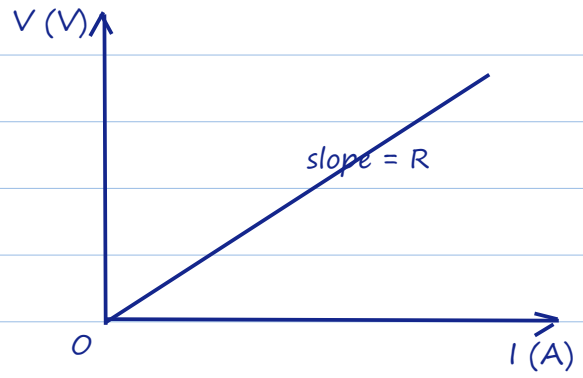
2.

a = (20 - 0)/5 = 4 m/s^2

distance s = v x t = 20 x 5 = 100 m

3.

Ohm's law states that V is directly proportional to I, V = IR, when temperature is constant.



R is found from the slope of the graph.

4.

Kinetic energy = energy of a moving body = 1/2 mv^2

Potential energy = energy stored due to height = mgh

PE at top = 2 x 10 x 20 = 400 J. All of it becomes KE at the bottom.

400 = 1/2 x 2 x v^2 so v^2 = 400, v = 20 m/s

Name: Rohan Verma | USN / Roll No: 1PH23CS027

Subject: Engineering Physics - IA 1 | Section: B

Q1

1st law - a body at rest remains at rest unless a force acts on it.

2nd law - F = ma, force gives acceleration to the body.

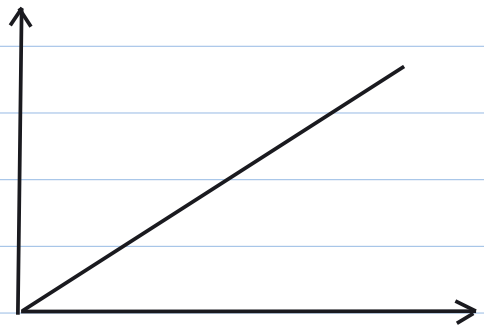
Q2

a = v/t = 20/5 = 4

s = 1/2 at^2 = 1/2 x 4 x 5 = 10 m

Q3

Ohms law: V = IR. Voltage is proportional to current.



Q4

KE = 1/2 mv^2 is energy of motion. PE = mgh.

mgh = 1/2 mv^2

v = 2gh = 2 x 10 x 20 = 400 m/s

Name: Ananya Iyer | USN / Roll No: 1PH23CS033

Subject: Engineering Physics - IA 1 | Section: B

Answer 1

Newtons laws are about force. First law is that force makes things move. Second law is heavier things fall faster. Third law action = reaction.

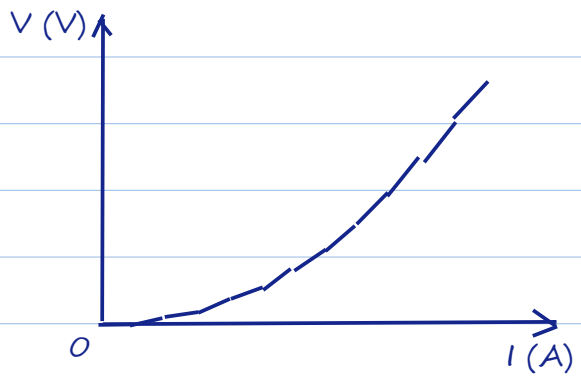
Answer 2

a = v x t = 20 x 5 = 100 m/s^2

s = 100 x 5 = 500 m

Answer 3

Ohm's law says resistance increases when current increases, so the graph is a curve.



Answer 4

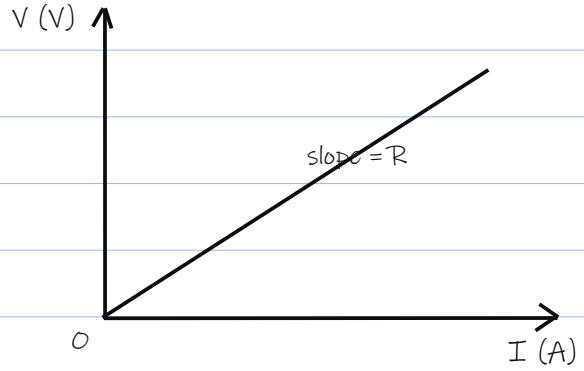
KE is energy stored in a body. PE is energy of moving. Speed = height / time, I don't know the time.

Name: Kabir Singh | USN / Roll No: 1PH23CS041

Subject: Engineering Physics - IA 1 | Section: A

Ans 3 (answering this first)

Ohm's law: The current through a conductor is directly proportional to the voltage across it, provided the temperature stays constant. $V = IR$.



Resistance = slope of the V-I line = V/I.

Ans 1

~~First law: F = ma~~

First law: a body stays at rest or in uniform motion unless an external force acts on it.

Second law: F = ma, rate of change of momentum is proportional to force.

Third law: action and reaction are equal and opposite, e.g. when we walk we push the ground back and the ground pushes us forward.

Ans 2

~~a = 20 x 5 = 100~~

$a = (v - u)/t = 20/5 = 4\ m/s^2$

for distance, $s = ut + 1/2\ at^2 = 1/2 \times 4 \times 25$

(Q4 not attempted - ran out of time)

Name: Meera Nair | USN / Roll No: 1PH23CS052

Subject: Engineering Physics - IA 1 | Section: C

1)

I law (inertia): no net external force means no change in the state of rest or uniform straight-line motion.

II law: $F = dp/dt = ma$.

III law: forces always occur in equal and opposite pairs acting on different bodies. Eg: a swimmer pushes water back and moves forward.

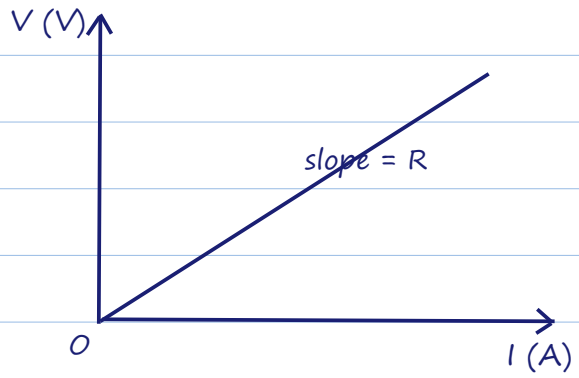
2)

(a) $a = (v - u)/t = 20/5 = 4\ m/s^2$

(b) $s = (u + v)/2 \times t = (0 + 20)/2 \times 5 = 50\ m$

3)

At constant temperature I is proportional to V, i.e. $V = IR$.



Straight line through origin; slope = $V/I = R$.

4)

KE = 1/2 mv^2 (due to motion), PE = mgh (due to position).

v^2 = u^2 + 2gh = 0 + 2 x 10 x 20 = 400, so v = 20 m/s.